Quantum Traffic Gridlock — Technical Report

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1. Mathematical Formulation & Exact QUBO Derivation

The traffic network consists of two connected intersections, `A` and `B`, optimized over four discrete 30-second time slots ($t \in \{0, 1, 2, 3\}$). The system is modeled using 8 binary decision variables $x_{I,t} \in \{0, 1\}$ ordered as `[A_slot1..4, B_slot1..4]`, where `1` denotes a North-South (NS) green phase and `0` denotes an East-West (EW) green phase. The objective is to minimize the official composite traffic cost:

$$\text{Cost}(x) = \text{Delay} + w_{\text{overflow}} \cdot \text{Overflow} + w_{\text{switch}} \cdot \text{Switches}$$

Rather than empirically tuning or hardcoding scenario parameters, our QUBO is derived analytically from first-principles traffic dynamics to ensure scenario-independent generalization.

1.1 Exact Switching Penalties

Phase switches between adjacent slots at each intersection are represented exactly as quadratic binary terms without approximation:

$$H_{\text{switch}} = w_{\text{switch}} \sum_{I \in \{A,B\}} \sum_{t=0}^{2} |x_{I,t+1} - x_{I,t}| = w_{\text{switch}} \sum_{I \in \{A,B\}} \sum_{t=0}^{2} \left(x_{I,t} + x_{I,t+1} - 2x_{I,t}x_{I,t+1}\right)$$

1.2 Effective Departure Capacity & Dynamic Link Coupling

To prevent unphysical negative queues without adding auxiliary slack qubits, the slot-0 departure capacity is bounded by initial arrivals: $C_{\text{eff},0} = \min(a_{app,0}, C)$, and $C_{\text{eff},s} = C$ for $s \geq 1$. For any approach, the cumulative unclipped queue at slot t is linear in the phase variables:

$$\tilde{q}_{app,t}(x) = \sum_{s=0}^{t} a_{app,s} + \sum_{s=0}^{t-1} \tau_{s+1}(x) - \sum_{s=0}^{t} C_{\text{eff},s}\, g_{app,s}(x)$$

where $g_{NS,s} = x_{I,s}$, $g_{EW,s} = 1 - x_{I,s}$, and directed $A \to B$ link transfers $\tau_{s+1} = \lfloor \alpha C + 0.5 \rfloor\, g_{A,\text{src},s}$ introduce off-diagonal bilinear couplings $Q_{i,j}$ between upstream green decisions at A (slot s) and downstream queue accumulation at B (slots $t > s$).

1.3 Quadratic Queue Barrier & Surrogate Fidelity

Because the exact overflow $\max(0, q_t - S)$ is piecewise-linear, we approximate combined delay and overflow via a calibrated quadratic barrier:

$$H_{\text{queue}} = \sum_{t=0}^{3} \sum_{app} \left(w_{\text{lin}} \tilde{q}_{app,t} + w_{\text{quad}} \tilde{q}^2_{app,t}\right), \quad w_{\text{lin}} = 1.0, \quad w_{\text{quad}} = \frac{w_{\text{overflow}}}{2SC}$$

Setting $w_{\text{quad}} = w_{\text{overflow}}/(2SC)$ ensures the marginal penalty $\partial H_{\text{queue}}/\partial \tilde{q}$ at the storage threshold $\tilde{q} = S$ scales proportionally with w_{overflow}. On Reference Scenario R, the QUBO achieves a **Pearson correlation of 0.9738** across all 256 states and **0.00% surrogate regret**, with its exact minimizer matching the true global optimum (`11101010`, $C^* = 72.0$). Across unseen scenarios (`heavy_ns.json`, `balanced.json`), surrogate regret remains $\leq 5\%$. Exact equivalence between the upper-triangular QUBO and Ising Hamiltonian ($x_i = (1 - Z_i)/2$) is verified across all 256 basis states to machine precision ($\leq 1.71 \times 10^{-13}$).

2. QAOA Implementation, Multi-Start Training & Ablations

2.1 Resource Adherence & Circuit Architecture

- **Qubits & Depth:** Exactly 8 qubits (limit ≤ 16) and depth $p = 3$ (limit $p \leq 4$), exported to OpenQASM 3.0 (`circuit.qasm`).
- **Evaluation & Shot Limits:** 150 COBYLA objective evaluations (limit ≤ 300) using exact statevector probabilities $\langle H_C \rangle = \sum_x p(x)E(x)$ during training, followed by **exactly 4,096 raw measurement shots** on `AerSimulator` (`counts.json`).
- **Endianness Decoding:** Qiskit little-endian bitstrings `[q7..q0]` are reversed (`bitstring[::-1]`) to map qubit i to schedule bit i.

2.2 Spectrally Normalized Multi-Start Optimization

To mitigate barren plateaus and local traps in COBYLA, we normalize the Ising coefficients by $h_{\text{scale}} = \max(1, 0.5(\max_i |h_i| + \max_{i<j} |J_{ij}|))$ and divide the 150-evaluation budget across three deterministic starts (50 evaluations each): (1) a linear ramp, (2) a spectrally normalized adiabatic ramp, and (3) a harmonic-seeded ramp.

2.3 Systematic Ablation Studies

Implemented in `extensions.py`, our ablations on Scenario R demonstrate:

1. **Circuit Depth ($p \in \{1, 2, 3, 4\}$):** Increasing depth from $p = 1$ to $p = 3$ steadily reduces expected QUBO energy and improves sampling lift; $p = 3$ offers the best accuracy-to-gate-count tradeoff before $p = 4$ experiences classical optimizer saturation under the evaluation cap.
2. **Penalty Scaling (quad_scale $\in \{0.25, 0.5, 1.0, 2.0, 4.0\}$):** The analytical calibration (scale $= 1.0$) maximizes Pearson correlation (0.9738) and yields 0.00% regret, whereas extreme under-scaling (0.25) under-penalizes storage overflow and over-scaling (4.0) distorts linear delay tradeoffs.
3. **Initialization Strategy:** Multi-start and linear/spectral ramps consistently outperform unscaled random initialization in both final expected cost E_Q and elite enrichment.

3. Baselines, Sampling Quality & Multi-Seed Robustness

3.1 Mandatory Baseline Comparison (Scenario R, Seed 42)

Baseline / Method	Schedule	Official Cost	Delay / Overflow / Switches	Notes
Exact Enumeration (C^*)	`11101010`	**72.00**	66 / 1 / 4	Global optimum over all 256 schedules (worst: `00001111` at 297.0).
Exact QUBO Minimizer	`11101010`	**72.00**	66 / 1 / 4	Surrogate ceiling; **0.00% regret** (QUBO energy: 89.26).
Per-Slot Greedy Control	`01111001`	**92.50**	67 / 6 / 3	Serves $\max(q_{NS}, q_{EW})$ online after arrivals (ties to NS).
Best Alternating	`10101010`	**116.00**	73 / 10 / 6	Best of 4 cyclic schedules (`10101010`, `10100101`, `01011010`, `01010101`).
Uniform Random (C_{rand})	N/A	**151.25**	N/A	Mean official cost across all 256 schedules.
QAOA Raw Sampling (Seed 42)	`11101010` (best)	**137.02** (E_Q)	66 / 1 / 4 (best)	Raw lift = **0.1795**, elite enrichment = **1.125x** (234 / 4,096 elite shots; `11101010` sampled 20 times).
Hybrid QAOA + Local Search	`11101010`	**72.00**	66 / 1 / 4	Separately labeled 1-bit-flip classical refinement from the best QAOA sample.

3.2 Five-Seed Robustness Evaluation (Scenario R)

Evaluating across 5 fixed seeds (`42`, `43`, `44`, `45`, `46`) yields positive lift on every seed, with **Mean Clipped Lift = 0.1545** and **Worst-Seed Clipped Lift = 0.0589**:

Seed	Mean Cost (E_Q)	Raw Lift	Clipped Lift $[0, 1]$	Elite Shots (of 4,096)	Elite Enrichment
42	137.0223	0.1795	0.1795	234	1.1250x
43	145.3016	0.0751	0.0751	234	1.1250x
44	133.2650	0.2269	0.2269	286	1.3750x
45	132.8557	0.2321	0.2321	207	0.9952x
46	146.5801	0.0589	0.0589	182	0.8750x

4. Innovation, Extensions & Limitations

4.1 Mathematical Extensions (`extensions.py`)

1. **Minimum-Green Phase Hold ($\tau_{\min} = 2$):** Rapid 1-slot phase oscillations (`1→0→1` or `0→1→0`) create traffic safety hazards. We add a 3-slot quadratic penalty $H_{\text{mg}} = \lambda_{\text{mg}} \sum_{I,t\in\{1,2\}} (x_{I,t} - x_{I,t-1}x_{I,t} - x_{I,t}x_{I,t+1} + x_{I,t-1}x_{I,t+1})$ with $\lambda_{\text{mg}} = 2.5$ within the same 8-qubit space, suppressing isolated 1-slot phase flips.
2. **Gate-Level Depolarizing Noise Analysis:** We simulate noisy execution on `AerSimulator` by injecting 2-qubit depolarizing errors $\epsilon_{2q} \in \{0.002, 0.005, 0.01, 0.02\}$ on `rzz`/`cx` gates and $\epsilon_{1q} = \epsilon_{2q}/10$ on `h`/`rx`/`rz` gates, quantifying graceful degradation of lift and elite enrichment on NISQ hardware.

4.2 Limitations

- **Unclipped Surrogate Queues:** To avoid adding auxiliary slack qubits per slot/approach, $\tilde{q}_{app,t}$ is modeled linearly without $\max(0, \cdot)$ clipping for $t \geq 1$. Under extremely sparse arrival regimes (e.g., consecutive zero arrivals), $\tilde{q}$ can under-estimate true queues if green phases starve.
- **Smooth Overflow Approximation:** The quadratic barrier $w_{\text{quad}}\tilde{q}^2$ penalizes moderate queues ($0 < \tilde{q} \leq S$) slightly more than the exact zero-below-storage hinge $\max(0, q - S)$, causing minor energy offset shifts while preserving state rankings.
- **Shallow-Depth QAOA Concentration:** At $p = 3$ with standard transverse-field mixers, QAOA reliably shifts probability mass toward low-cost schedules (positive lift and global optimum sampling) but retains broad support across the 256-state space compared to classical exact enumeration.